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10 CONCEPTUAL Q&A PROGRAMMING LANGUAGES · PRACTICE WORKSHEET

CONCEPTUAL SERIES
C

Name: Cohort/Batch: Date: Score:

C PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Programming Languages

TOTAL: 10 QUESTIONS

Q1 What are the primary design goals of C? — Threat Model

Q2 How does C handle type checking: static or dynamic? — Architecture

Q3 How does C manage memory? — Configuration

Q4 What is C's concurrency model? — Hardening

Q5 How are packages and dependencies managed in C? — SSO / IdP

Q6 How does C implement object-oriented or functional programming? — Observability

Q7 How does C handle errors? — Lifecycle

Q8 What testing tools are commonly used in C? — Compliance

Q9 What makes C well-suited for its target domain? — Testing

Q10 What are common performance pitfalls in C? — Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 C was designed with specific priorities around
Mitigation

C was designed with specific priorities around performance, safety, or developer productivity depending on its domain. These goals shape its syntax, type system, and standard library choices.

A2 C uses its type system to catch
Primitives

C uses its type system to catch errors at compile time (static) or at runtime (dynamic). This affects refactoring safety, tooling support, and the kinds of bugs that reach production.

A3 C uses one of three approaches: manual
Hardening

C uses one of three approaches: manual allocation/deallocation, a garbage collector that periodically reclaims unreachable objects, or automatic reference counting. Each has different latency and throughput tradeoffs.

A4 C supports concurrency through threads, coroutines,
Gotchas

C supports concurrency through threads, coroutines, async/await, or an actor model. The choice determines how I/O-bound and CPU-bound workloads are scaled and how shared state is safely accessed.

A5 C uses a package manager and a
Federation

C uses a package manager and a manifest file to declare dependencies and version constraints. A lock file pins exact versions to ensure reproducible builds across environments.

A6 C supports classes and inheritance for OOP,
Observability

C supports classes and inheritance for OOP, or first-class functions and immutability for FP, or both. Many modern C programs blend both paradigms depending on the problem.

A7 C surfaces errors via exceptions, Result/Either types,
Automation

C surfaces errors via exceptions, Result/Either types, or error return values. The choice impacts whether errors are checked at compile time and how calling code is structured.

A8 C has a standard or widely adopted
Governance

C has a standard or widely adopted test runner and assertion library. Tests are typically organized into unit, integration, and end-to-end layers with coverage reporting built in or via plugins.

A9 C excels in its domain due to
Assessment

C excels in its domain due to a combination of performance characteristics, ecosystem maturity, language ergonomics, and community support that makes common tasks simple.

A10 Typical performance issues in C include excessive
Optimization

Typical performance issues in C include excessive allocations, blocking the main thread or event loop, O(n^2) data structures, and missing caching. Profiling and benchmarking tools are available to diagnose them.